

COURSE TITLE : POLYMER SCIENCE
COURSE CODE : 3092
COURSE CATEGORY : B
PERIODS/WEEK : 4
PERIODS/SEMESTER: 60/3
CREDITS : 4

TIME SCHEDULE

Module	Topics	Period
1	State of phase and aggregate of polymers and polymer identification	15
2	Molecular weight of polymers	15
3	Stereo regularity of polymers	15
4	Polymer reactions and polymer modifications	15
TOTAL		60

COURSE OTCOME

Sl.	G.O	Student will be able to
1	1	To understand the Tg , crystallinity and physical and chemical methods of polymer identification
	2	To understand the concept of molecular weight of polymers
2	1	To know the stereo regularity of polymers
3	1	To comprehend Polymer reactions and polymer modifications

SPECIFIC OUTCOME

MODULE I STATE OF PHASE AND AGGREGATE OF POLYMERS AND POLYMER IDENTIFICATION

1.1.0 To understand the Tg , crystallinity and physical and chemical methods of polymer identification

- 1.1.1. Define the terms Glass transition temperature (Tg), Melting temperature(Tm) and Flow temperature(Tf)
- 1.1.2. Distinguish between amorphous and crystalline polymers with examples.
- 1.1.3. Define crystallinity and crystallisability of polymers.
- 1.1.4. Explain the different factors affecting Tg of polymers.
- 1.1.5. Describe the method of determining Tg by dilatometry.
- 1.1.6. Explain the X-Ray diffraction method for determining crystallinity of polymers.
- 1.1.7. Compare Elastomers, Plastics and Fibres in terms of Tg and crystallinity.
- 1.1.8. Discuss the importance of polymer identification.

- 1.1.9. Describe the detailed scheme for identification of polymers by chemical method.
- 1.1.10. Explain the physical methods to analyse polymers by TGA, DSC and IR spectroscopy.

MODULE II MOLECULAR WEIGHT OF POLYMERS.

2.1.0 To understand the concept of molecular weight of polymers.

- 2.1.1. State the concept of average molecular weight of polymers.
- 2.1.2. Distinguish between the molecular weight of polymers and simple molecules.
- 2.1.3. List various molecular weights of polymers with corresponding equations
- 2.1.4. Derive equations for M_n and M_w .
- 2.1.5. Define polydispersity and molecular weight distribution.
- 2.1.6. Explain the techniques of determination of M_n by End group analysis, Ebulliometry, cryoscopy, Osmometry.
- 2.1.7. Explain the techniques of determination of molecular weight by Light scattering and Gel permeation chromatography.
- 2.1.8. Explain the techniques of determination of Viscosity average molecular weight.

MODULE III STEREOREGULARITY OF POLYMERS

- 3.1.0 To know the stereo regularity of polymers
- 3.1.1. State the general remarks on polymer microstructure.
- 3.1.2. Explain the microstructures based on chemical and geometrical structure
- 3.1.3. Define stereo regular polymers.
- 3.1.4. Explain stereo regularity of polymers -optical and geometrical isomerism with examples
- 3.1.5. State the importance Z-N catalyst in stereo regularity of polymers.
- 3.1.6. List different combinations of ZN catalyst.
- 3.1.7. Explain the mechanism of formation of stereo regularity in polymers by using Z-N catalyst.
- 3.1.8. Define tacticity of polymers, isotactic, syndiotactic and atactic with respect to polypropylene.
- 3.1.9. Explain the microstructure- cis, trans and vinyl- of BR, IR, CR.

MODULE IV POLYMER REACTIONS AND POLYMER MODIFICATIONS

4.1.0 To comprehend Polymer reactions and polymer modifications

- 4.1.1. Identify the differences between reactions of polymers and simple molecules
- 4.1.2 .Explain the effect of functional groups in a polymer such as hydroxyl, carboxylic and aldehydic groups
- 4.1.3. Explain the various reactions –Hydrolysis, Acidolysis, Aminolysis, Hydrogenation Addition, Substitution and Cross-linking reactions
- 4.1.4. Classify the physical and chemical methods of polymer modification
- 4.1.5. Explain the various physical methods of polymer modification such as blends, Alloys and composites
- 4.1.6. Explain the various chemical methods of polymer modification such as co-polymerisation, grafting, cross-linking and interpenetrating network

CONTENT DETAILS

MODULE I

State of phase and aggregates of polymers and polymer identification

Glass transition temperature (T_g)-Melting temperature (T_m) –Flow temperature (T_f). Crystalline polymer –Amorphous polymer –Crystallinity Crystallisability -factors affecting T_g – Dilatometry-determination of T_g-X-Ray diffraction method for determination of crystallinity – Comparison of Elastomers ,plastics and fibers in terms of T_g and crystallinity .Polymer identification –chemical and physical methods . TGA –DSC and IR spectroscopy.

MODULE II

Molecular weight of polymers

Average molecular weight concept –different molecular weight of polymers –Distinguish molecular weight of polymers and polymers-Different molecular weights of polymers –M_n, M_w, M_v and M_z-Derive equations for M_n and M_w. Polydispersity and molecular weight distribution. Determination of M_n and M_w- End group analysis –Ebulliometry –Cryoscopy –Osmometry –Light scattering and Gel permeation chromatography –Viscometry methods .

MODULE III

Stereo regularity of polymers

Polymer microstructure –chemical and Geometrical structure –Stereo regular polymer –Optical and geometrical isomerism. Ziegler Natta catalyst- typical combination- formation. Tacticity- isotactic, syndiotactic, atactic. Microstructure- BR, IR,CR- Cis, Trans, 1,2 addition, 3,4 addition and tactic structures.

MODULE IV

Polymer reactions and polymer modifications

Differences between reactions of polymers and simple molecules –effect of functional groups in polymers – Hydroxyl- carboxylic –aldehydic groups. Hydrolysis –Acidolysis –Aminolysis –Hydrogenation – Addition –Substitution – cross linking reactions . Polymer modification —Physical methods .Blends – Alloys Composites – co polymerization –Grafting –cross linking –IPN -Chemical methods

REFERENCES

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|-----------------------------------|--|
| 1. Polymer science | -V.R.Gowarikar (Publisher – New Age International (P) Ltd, New Delhi- 2005 Edition |
| 2. Text Book of Polymer Science | -Billmeyer.J.R (John Wiley & Sons (Aia) PTE Ltd, Singapore, 2008 Edition |
| 3. Introductory Polymer Chemistry | - G.S.Misra (Publisher – New Age International (P) Ltd, New Delhi- 2005 Edition |
| 4. Principles of Polymer Science | -P.Bahadur, N.V.Sastri (Publisher – Alpha Science International Ltd, 2005 Edition |